

IBM Watson OpenScale

demand_forecasting_lgbm Evaluation Report

January 28, 2026

Overview

Deployed model:

demand-forecasting-ml

Total test(s) failed

1

Report Details

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Report generated by: Nayef Abou Tayoun (Nayef.Abou.Tayoun@ibm.com)
Report generated on: January 28, 2026 21:30:21 UTC

Model Details

Deployment ID: 51e0d9ed-aaa9-4e48-a184-bd83a81f1157
Model name: demand_forecasting_lgbm
Model ID: a41fe4b6-02e4-459f-95f6-9d3cd6cf870a
Data type: Numeric
Algorithm type: Regression
Number of explanations: 2

Training data details

Storage location: dataset
Label column: target
Deployment prediction: prediction
Training features: PART_NUM, STOCKING_PCT, PART_SALES_QTY, PART_SALES_AMT, PART_SHIP_QTY, PART_SHIP_AMT, PART_DEMAND, PART_CLASS_CDE, PART_SALES_QTY_lag1, PART_SALES_QTY_lag2, PART_SALES_QTY_lag3, PART_SALES_QTY_lag4, PART_SALES_QTY_lag5, PART_SALES_QTY_lag6, rmean_PART_SALES_QTY_3, month, quarter, year, age_month

Metrics

Metric details

Summary

Deployed model	Model ID	Test data set
demand-forecasting-ml	a41fe4b6-02e4-459f-95f6-9d3cd6cf870a	test_data.csv

Metric

Global explanation

Alerts

0

GREEN

Summary

Explanation method:	Lime
Number of metrics:	1
Alerts:	0

Statistics

Global explanation stability

Status:	GREEN
Score:	0.999
Threshold:	0.800

Feature influence

PART_DEMAND:	2049.170
age_month:	258.626
PART_CLASS_CDE:	175.402
PART_NUM:	169.920
PART_SALES_QTY_lag5:	165.617
month:	152.545
PART_SHIP_QTY:	107.237
PART_SALES_QTY_lag1:	105.652
PART_SALES_QTY:	101.704
PART_SALES_QTY_lag3:	97.575
SUM of 9 other features:	263.76800000000003

Properties

Evaluated records count:	500
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Metric

Quality

Alerts

0

GREEN

Metrics

Summary

Number of metrics:	9
Alerts:	0

Statistics

Spearman

Status:	GREEN
Score:	0.77
Threshold:	0.60

Proportion explained variance

Status:	GREEN
Score:	37151.56
Threshold:	--

Pearson

Status:	GREEN
Score:	0.81
Threshold:	0.80

Smape

Status:	GREEN
Score:	0.86
Threshold:	--

Root of mean squared error

Status:	GREEN
Score:	159.98
Threshold:	2454.00

Mean absolute error

Status:	GREEN
Score:	124.59
Threshold:	--

Mean squared error

Status:	GREEN
Score:	25594.48
Threshold:	--

Mape

Status:	GREEN
Score:	3.96
Threshold:	--

R squared

Status:	GREEN
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Metrics

Score: 0.51
Threshold: - -

Properties

Minimum sample size: 1000
Evaluated records count: 1000

Metric

Drift v2

Alerts

1

RED BREACH

Summary

Number of metrics: 2
Alerts: 1

Confidence drift

Status: **RED BREACH**
Score: 0.32
Threshold: 0.20

Feature drift

Status: **GREEN**
Score: 0.20
Threshold: 0.20

Properties

Evaluated records count: 1000

Test summary

Tests passed

2

Tests failed

1

Number of evaluated records

1000

Appendix

Quality Measures

Area under ROC
Area under PR
Accuracy
True positive rate (TPR)
False positive rate (FPR)
Recall
Precision
F1-measure
Logarithmic loss

Fairness measures

Fairness

Drift v2 measures

Output drift
Model quality drift
Feature drift
Input metadata drift
Prediction drift
Embedding drift
Output metadata drift

Appendix

Quality measures

Area under ROC

The Area under ROC is plotted parametrically as the [True positive rate](#) versus the [False positive rate](#) with respect to a threshold T .

Area under PR

Area under Precision Recall gives the total for both [Precision + Recall](#). Precision (P) is defined as the number of true positives (Tp) over the number of true positives plus the number of false positives (Fp)

Formula

$$\text{Precision} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false positives})}$$

Recall (R) is defined as the number of true positives (Tp) over the number of true positives plus the number of false negatives (Fn).

$$\text{Recall} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

Appendix

Quality measures

Accuracy

Base accuracy is calculated from the training data. It is the percentage of predictions that the model got correct when tested against the training data.

True positive rate (TPR)

The True positive rate is calculated by the following formula:

Formula

$$\text{TPR} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

False positive rate (FPR)

The false positive rate is calculated as the total number of false positives divided by the number of false positives and the number of true negatives.

$$\text{FPR} = \frac{\text{number of false positives}}{(\text{number of false positives} + \text{number of true negatives})}$$

Appendix

Quality measures

Recall

Recall (R) is defined as the number of true positives (Tp) over the number of true positives plus the number of false negatives (Fn).

Formula

$$\text{Recall} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

Precision

Precision (P) is defined as the number of true positives (Tp) over the number of true positives plus the number of false positives (Fp).

Formula

$$\text{Precision} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false positives})}$$

Appendix

Quality measures

F1-Measure

The F1-Measure is the weighted harmonic average, or mean, of precision and recall.

Formula

$$F1 = 2 * \frac{(\text{precision} * \text{recall})}{(\text{precision} + \text{recall})}$$

Logarithmic loss

For a binary model, Logarithmic loss is calculated by using the following formula:

Formula

$$-(y \log(p) + (1-y) \log(1-p))$$

where p = true label and y = predicted probability

For a multi-class model, Logarithmic loss is calculated by using the following formula:

$$-\sum_{c=1}^M Y_{o,c} \log(P_{o,c})$$

where $M > 2$, p = true label, and y = predicted probability

Appendix

Fairness measures

Fairness

The fairness metric used in Watson OpenScale is disparate impact, which is a measure of how the rate at which an unprivileged group receives a certain outcome or result compares with the rate at which a privileged group receives that same outcome or result.

Formula

$$\text{Disparate impact} = \frac{(\text{num_positives(privileged=False)} / \text{num_instance(privileged=False)})}{(\text{num_positives(privileged=True)} / \text{num_instance(privileged=True)})}$$

Appendix

Drift v2 measures

Output drift

Measures the change in the model confidence distribution.

Model quality drift

Measures the drop in accuracy by comparing the estimated runtime accuracy to the training accuracy. Ground truth labels are not required.

Feature drift

Measures the change in value distribution for important features.

Input Metadata drift

Measures the change in the distribution of various input metadata e.g. input token count, input character count, etc.

Output Metadata drift

Measures the change in the distribution of various output metadata e.g. output token count, output character count, etc.

Embedding drift

Measures the percentage of transactions marked as outliers based on cosine and euclidean distance based on the embeddings of the input, output, individual variables.